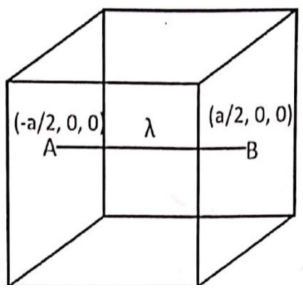
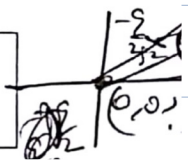
(I) A wire AB of uniform charge density λ is placed in a cubic box of sides a as shown in the figure below. The volume charge density within the box is $\rho = \lambda\,\delta(y)\delta(z)$.

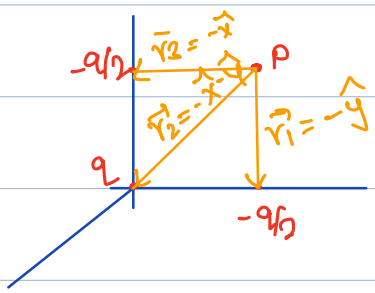
True ✓



8. Consider the charge configuration (x, y, z): q at (0,0,0), $-\frac{q}{2}$ at (1,0,0), and $-\frac{q}{2}$ at (0,1,0). The dipolar potential at (1,1,0) is given by

$\frac{-q}{8\sqrt{2}\pi\varepsilon_0}$ ✓ −1 ✓





$\vec{P} = \sum q_i \vec{r}_i$

$= -q/2(-\hat{y}) - q/2(-\hat{x}) + q(-\hat{x}-\hat{y})$

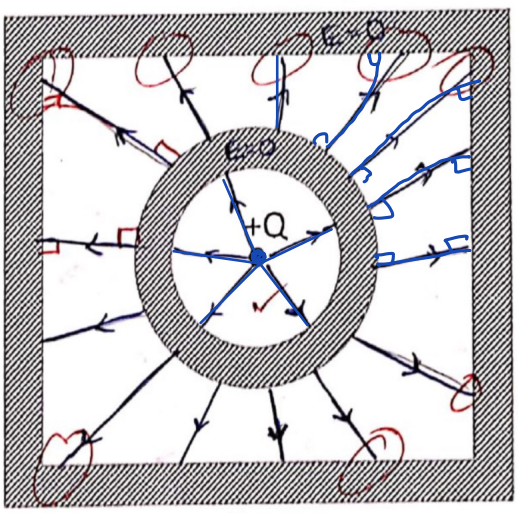
$\vec{P} = -q/2(\hat{x}+\hat{y})$

$\hat{r} = \frac{\hat{x}+\hat{y}}{\sqrt{2}}$

$V = \frac{1}{4\pi\varepsilon_0}\frac{\vec{P}\cdot\hat{r}}{r^2}$

$= \frac{1}{4\pi\varepsilon_0(2)}\frac{-q}{\sqrt{2}} = \frac{-q}{8\sqrt{2}\pi\varepsilon_0}$

2. A point charge $+Q$ is placed at the center of a conducting spherical shell, placed inside a cubic box with conducting walls as shown in the figure below. Sketch the field lines everywhere within the cube including the spherical shell. [*2 marks*]



Don't overwrite

5. An ideal dipole of magnitude p_0 is lying in the xy-plane oriented at an angle of 45^0 with respect to the x-axis. This dipole is placed in an electric field given by [*1 mark*]

$$\vec{E} = (\alpha x^2 + \beta x)\hat{x},$$

where α and β are constants of appropriate dimensions. The force on the dipole is

$$\vec{F} = (\vec{p}\cdot\vec{\nabla})\vec{E} = \frac{p_0}{\sqrt{2}}(2\alpha x + \beta)\hat{x}$$

$$\vec{F} = (\vec{p}\cdot\vec{\nabla})\vec{E} = p_x\frac{dE_x}{dx} + p_y\frac{dE_y}{dy} = p_0/\sqrt{2}(2\alpha x+\beta)\hat{x}$$

4. An ideal dipole with moment $\vec{p} = p_0\hat{z}$ is placed at the origin. The electric field due to this dipole at a point lying on the xy plane, and at a distance s from the z axis, is [*1 mark*]

$$\vec{E} = \frac{1}{4\pi\varepsilon_0 s^3}\left[3(\vec{p}\cdot\hat{r})\hat{r} - \vec{p}\right] = \frac{-p_0}{4\pi\varepsilon_0 s^3}\hat{p} = -\frac{p_0\hat{x}}{4\pi\varepsilon_0 s^3} + \frac{p_0}{4\pi\varepsilon_0 s^3}$$

The flux of the electric field through the upper hemisphere ($z \geq 0$) of a sphere of radius R centred at the origin, is [*1 mark*]

$$\phi = \frac{p_0}{2\varepsilon_0 R}$$

$$dS = R^2\sin\theta\, d\theta\, d\phi$$

$$\vec{E}\cdot d\vec{S} = \int \frac{p_0}{4\pi\varepsilon_0 R^3}\times R^2\sin\theta\cos\theta\, d\theta\, d\phi$$

$$= \frac{p_0}{} \int \sin 2\theta\, d\theta$$

6. An infinite slab of uniform volume charge density ρ is kept in the *xy*- plane between $z = -a$ and $z = a$. Which of the following statement(s) are correct. [*2 marks*]

(A) The field everywhere outside the slab is zero.

(B) The field at the origin is $\frac{\rho}{3\epsilon_0}$.

(C) The field inside the slab is proportional to the distance from the *xy* plane.

(D) The field everywhere inside the slab is directed along the positive *z*-axis.



9. Consider a uniformly charged body (see the figure below) with charge density ρ, whose shape is defined by

[*3 marks*]

$$\frac{x^2}{a^2} + \frac{y^2}{a^2} + \frac{z^2}{b^2} \leq 1.$$

Find the quadrupole moment components Q_{zz} and Q_{xy} about the origin O.

